

ANX-PR/CL/001-01

LEARNING GUIDE

SUBJECT

103000752 - Software Project Management

DEGREE PROGRAMME

10AK - Master Universitario En Software Y Sistemas

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

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1. Description

1.1. Subject details

Name of the subject	103000752 - Software Project Management
No of credits	4 ECTS
Type	Optional/elective
Academic year of the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	10AK - Master Universitario en Software y Sistemas
Centre	10 - E.T.S. De Ingenieros Informáticos
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Ana Maria Moreno Sanchez-Capuchino (Subject coordinator)	5102	anamaria.moreno@upm.es	M - 15:00 - 21:00
Tomas San Feliu Gilabert	D 5105	tomas.sanfeliu@upm.es	M - 10:00 - 12:00 W - 10:00 - 12:00

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Skills and learning outcomes *

3.1. Skills to be learned

CEM4 - Analizar y evaluar los diferentes paradigmas y enfoques de ingeniería de construcción y gestión de sistemas basados en software.

CG13 - Apreciación de los límites del conocimiento actual y de la aplicación práctica de la tecnología más reciente.

CG4 - Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CG8 - Planteamiento y resolución de problemas también en áreas nuevas y emergentes de su disciplina

CG9 - Aplicación de los métodos de resolución de problemas más recientes o innovadores y que puedan implicar el uso de otras disciplinas

CGI23 - Capacidad de leer y comprender publicaciones dentro de su ámbito de estudio/investigación, así como su catalogación y valor científico

3.2. Learning outcomes

RA131 - Development of a business case for a software project

RA132 - Development of a project plan at different granularity levels

RA130 - Identify the elements of a risk management plan and its rationale

RA129 - Communication Skills

RA133 - Development of a communication plan for a software project

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

4. Brief description of the subject and syllabus

4.1. Brief description of the subject

This subject covers the different activities to be performed by a software project manager to successfully drive a software project. For that aim, both workshops and theoretical lectures will be provided.

4.2. Syllabus

1. Introduction to Software Project Management
 - 1.1. Psychological Model of Software Engineers
 - 1.2. Leadership versus Management
 - 1.3. The Basic Functions of Project Management
 - 1.4. Developing the Business Case
2. Developing and Motivating the Project Team
 - 2.1. What it Takes for a Team to be Effective
 - 2.2. The Basics of Personalities
 - 2.3. Motivating Software Engineers
3. Strategic Approaches to Project Planning
 - 3.1. The Balanced Scorecard (BSC)
 - 3.2. Using SWOT to Develop the BSC
 - 3.3. Tracking Progress
4. Estimating Project Size, Cost, Schedule
5. Project Risk Management
 - 5.1. What Risk is/is not
 - 5.2. Strategies for Mitigating Risk
 - 5.3. Risk Management Methods
6. Tracking the Progress of a Software Project
 - 6.1. Earned Value Management

6.2. Time Value

5. Schedule

5.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	Chapter 1. Introduction to Software Project Management Duration: 02:00 Lecture			
2	Software Project Management Model (Lego) Duration: 02:00 Gamification			
3	Chapter 2: Managing and Motivating your Team Duration: 02:00 Lecture			Partial Presentation Group presentation Progressive assessment Presential Duration: 02:00
4	Chapter 2. Know your team Duration: 02:00 Cooperative activities			
5	Chapter 3. Strategic Approaches to Project Planning Duration: 02:00 Lecture			
6	Chapter 4. Estimating and Sizing your Project Duration: 02:00 Lecture			Partial Presentation Group presentation Progressive assessment Presential Duration: 02:00
7	Building Blocks Workshop Duration: 02:00 Gamification			
8	SPM Complementary Knowledge Duration: 02:00 Cooperative activities			
9	Chapter 5. Risk Management Duration: 02:00 Lecture			
10	Chapter 5. Risk Management Duration: 02:00 Cooperative activities			Partial Presentation Group presentation Progressive assessment Presential Duration: 02:00
11	Chapter 6. Project Monitoring Duration: 02:00 Cooperative activities			

12	Chapter 6. Project Monitoring Duration: 02:00 Cooperative activities			Partial Presentation Group presentation Progressive assessment Presential Duration: 02:00
13	Soft Skills Workshop Duration: 02:00 Cooperative activities			
14	Software Project Management Metamodel (Iego) Duration: 02:00 Gamification			
15	Project Term Delivery Duration: 02:00 Additional activities			Final Report Group work Progressive assessment Not Presential Duration: 00:00 Final presentation Group work Progressive assessment Presential Duration: 02:00 Active Participation of Students Other assessment Progressive assessment Not Presential Duration: 00:00
16	Industrial Seminar Duration: 02:00 Additional activities			
17				Content of Final Report Group work Global examination Not Presential Duration: 00:00 Active Participation of Students Other assessment Global examination Not Presential Duration: 00:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

6. Activities and assessment criteria

6.1. Assessment activities

6.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
3	Partial Presentation	Group presentation	Face-to-face	02:00	10%	0 / 10	CG4 CG8 CG9 CG13 CGI23 CEM4
6	Partial Presentation	Group presentation	Face-to-face	02:00	10%	0 / 10	CG4 CG8 CG9 CG13 CGI23 CEM4
10	Partial Presentation	Group presentation	Face-to-face	02:00	10%	0 / 10	CG4 CG8 CG9 CG13 CGI23 CEM4
12	Partial Presentation	Group presentation	Face-to-face	02:00	10%	0 / 10	CG4 CG8 CG9 CG13 CGI23 CEM4
15	Final Report	Group work	No Presential	00:00	40%	5 / 10	CG4 CG8 CG9 CG13 CGI23 CEM4
15	Final presentation	Group work	Face-to-face	02:00	10%	5 / 10	CG4 CG8 CG9 CG13 CGI23 CEM4

15	Active Participation of Students	Other assessment	No Presential	00:00	10%	0 / 10	CG4 CG8 CG9 CG13 CEM4
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6.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
17	Content of Final Report	Group work	No Presential	00:00	85%	5 / 10	CG4 CG8 CG9 CG13 CGI23 CEM4
17	Active Participation of Students	Other assessment	No Presential	00:00	15%	0 / 10	CG4 CG8 CG9 CG13 CEM4

6.1.3. Referred (re-sit) examination

Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
Content of Final Project Report	Individual work	Face-to-face	00:00	85%	5 / 10	CG4 CG8 CG9 CG13 CGI23 CEM4
Active participation of students	Other assessment	Face-to-face	00:00	15%	0 / 10	CG4 CG8 CG9 CG13

6.2. Assessment criteria

During the progressive evaluation, the final grade of students will be calculated according to their performance in the reports to be done and their class participation.

- Active participation of students (10%)
- Content of the reports (40%)
- Presentation of the report (50% - 10% each presentation)

Students must get a minimum of 5 points (over 10) in the assessment of the content of the report in order to pass the matter.

Students must get a minimum of 5 points (over 10) as final grade in order to pass the matter.

During the global and extraordinary evaluation, the final grade of students will be calculated according to their performance in the report to be done and their class participation. If students have attended to the lecturers during the course, they will be evaluated accordingly in the "Active participation" activity. If not, their grade in the "Active participation" activity will be 0

- Active participation of students (15%)
- Content of the reports (85%)

Students must get a minimum of 5 points (over 10) in the assessment of the report in order to pass the matter.

7. Teaching resources

7.1. Teaching resources for the subject

Name	Type	Notes
Bibliography	Bibliography	Managing Software Projects on the Edge of Chaos: from Antipatterns to Success. Kindle eBook, by Lawrence Peters, (second edition) 2024 .